

Effectiveness of Transcutaneous Auricular Vagal Nerve Stimulation on Alleviating Postoperative Pain Following Thoracoscopic Lobectomy

A Participant- and Assessor-blinded, Randomized Controlled Trial

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Objectives: This study aimed to investigate the effectiveness of transcutaneous auricular vagal nerve stimulation (taVNS) in alleviating postoperative pain following thoracoscopic lobectomy.

Methods: In this randomized controlled trial, 70 adult patients were randomly assigned in a 1:1 ratio to receive either active or sham taVNS. Stimulation was initiated in the preoperative room, maintained throughout surgery, and discontinued after extubation in the postanesthesia care unit, at which point the device was removed. Additional 2-hour sessions were conducted each morning on postoperative days 1 and 2. Patient-controlled intravenous analgesia and serratus anterior plane block were provided to all the patients. The primary outcome was sufentanil consumption during the first 24 hours after surgery.

Results: A total of 67 participants completed the study. Sufentanil consumption showed no statistically significant difference between

the active group ($33 \pm 13 \mu\text{g}$) and the sham group ($32 \pm 9 \mu\text{g}$) (mean difference = 1.20, 95% CI = -4.23 to 6.63, P -value = 0.661). Compared with the sham group, taVNS produced a statistically significant reduction in the numerical rating scale for resting pain (P -value = 0.001) and deep breathing pain (P -value < 0.001) postoperatively. taVNS also significantly reduced the incidence of rebound pain (58% vs. 18%; P -value = 0.001) and chronic postsurgical pain at 3 months (27% vs. 6%; P -value = 0.018).

Discussion: taVNS did not reduce postoperative opioid consumption but alleviated acute postoperative pain, lowered the incidence of rebound pain and chronic postsurgical pain. These preliminary findings have uncertain clinical significance and require further investigation.

Clinical Trial Registration: Chinese Clinical Trial Registry (ChiCTR2400081062).

Key Words: video-assisted thoracic surgery, rebound pain, chronic postsurgical pain, transcutaneous auricular vagal nerve stimulation, taVNS

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KEY MESSAGES

What is already known on this topic

Thoracoscopic surgery is often associated with acute pain, rebound pain, and chronic postsurgical pain (CPSP). Transcutaneous auricular vagal nerve stimulation (taVNS) has shown promise in managing perioperative pain, but its specific effectiveness in thoracoscopic lobectomy has not been well established.

What this study adds

While taVNS did not significantly reduce opioid consumption, it significantly alleviated acute postoperative pain and reduced the incidence of rebound pain following a nerve block. Moreover, taVNS was associated with a notable decrease in the prevalence of CPSP at 3 months postsurgery.

INTRODUCTION

Despite the widespread application of multimodal analgesia techniques, minimally invasive thoracic surgery can still result in 3 types of unpleasant pain experiences: acute postoperative pain, rebound pain, and chronic postsurgical pain (CPSP). The degree of acute postoperative pain can be moderate to severe,¹ with significant demand for opioids within the first 24 hours postoperatively (morphine $42 \pm 12 \text{ mg}$).² Regional blockade, a key component of

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multimodal analgesia, is widely used in managing pain after video-assisted thoracoscopic surgery (VATS);³ however, the ensuing rebound pain when the block effect subsides is clinically relevant, with some requiring up to 97 mg of morphine within 10 hours.⁴ Furthermore, the incidence of CPSP remains high, reaching 25% to 44%.^{3,5–7}

Recent studies indicate that vagal activity is essential for preserving physical function, reducing pain, minimizing postoperative complications, and expediting recovery.^{8,9} Perioperatively, factors such as anesthesia and systemic inflammation can disrupt autonomic function,^{10–12} leading to sympathetic overactivity and reduced vagal tone,^{8,13,14} potentially worsening pain. Transcutaneous auricular vagal nerve stimulation (taVNS) modulates vagal tone by stimulating the auricular branch of the vagus nerve in the cymba conchae^{15,16} and shows promise in perioperative pain management by enhancing the release of endogenous acetylcholine.^{17–19}

While taVNS, as a noninvasive technique, holds significant potential for managing perioperative pain, the existing evidence from randomized controlled trials is insufficient to conclusively determine its effectiveness. This study aimed to investigate the effectiveness of taVNS in alleviating postoperative pain following thoracoscopic lobectomy.

taVNS may offer a promising nonopioid adjunct for managing acute postoperative pain and reducing the risk of CPSP in patients undergoing thoracoscopic lobectomy. The results of this study could help shape future pain management strategies and stimulate further research on the role of taVNS in thoracic surgery.

METHODS

Study Design and Participants

This single-center, participant- and assessor-blinded, randomized controlled trial was conducted at Sir Run Run Shaw Hospital, Zhejiang University School of Medicine, China from March 1, 2024, to July 9, 2024. The study was approved by the Sir Run Run Shaw Hospital Research Ethics Committee (Chair: Professor Limin Liu) (Approval number: 20240030) and was officially registered with the Chinese Clinical Trial Registry on February 21, 2024 (ChiCTR2400081062). The first patient was enrolled on March 26, 2024. The study followed the CONSORT 2010 guidelines (see Supplementary CONSORT 2010 Checklist, Supplemental Digital Content 1, <http://links.lww.com/CJP/B219>). All patients provided written informed consent before enrollment.

Inclusion Criteria

Eligible participants were with the American Society of Anesthesiologists (ASA) physical status I to III, aged 18 to 85 years, diagnosed with lung cancer at our center, and scheduled to undergo elective video-assisted thoracoscopic lobectomy under general anesthesia.

Exclusion Criteria

Exclusion criteria were (1) arrhythmias requiring intervention; (2) history of thoracic surgery; (3) neuromuscular disorders or dermatitis of the left ear; (4) substance abuse, including patients taking prescribed opioids or those with opioid use disorder (OUD); (5) inability to participate in assessments; (6) seizures or syncope within the past 5 years; (7) previous transient ischemic attacks or

cerebrovascular events; (8) vagus nerve stimulation or acupuncture within the past month; (9) concurrent participation in other clinical trials; (10) vulnerable populations, including individuals with mental illness, cognitive impairment, or pregnant women; (11) patients unable to provide informed consent.

Randomization and Masking

The research coordinator used computer-generated random numbers to allocate patients in a 1:1 ratio to either the active or sham stimulation group. The randomization results were sealed in opaque envelopes. Upon arrival at the preoperative preparation room, patients were screened according to the inclusion and exclusion criteria. A non-blinded individual opened the envelope to disclose the group assignment. A trained, nonblinded individual delivered either active or sham stimulation.

This study used a participant- and assessor-blinded design, in which only the individuals responsible for randomization and the intervention provider knew the treatment allocations. Throughout the study, participants, follow-up staff, and data collection and analysis staff remained blinded to the group assignments.

Intervention

The auricular vagus nerve stimulator (tVNS501), produced by Ruishen An Medical Devices Co., Ltd. (Changzhou, Jiangsu Province, China) was used in this study (Fig. 1). Only the left earpiece, equipped with embedded metal electrodes for targeted stimulation of the cymba concha, was used. Although the commercial device includes a right earpiece, it lacks stimulation capability and was neither worn nor activated during the trial.

The stimulator operator initiated either active or sham taVNS for all patients in the preoperative preparation room. This stimulation was maintained throughout surgery and discontinued after extubation in the postanesthesia care unit (PACU), at which point the device was removed. Two additional 2-hour sessions were conducted each morning on postoperative days 1 and 2. The devices were labeled to ensure that each patient received the same device for each session.

To determine the optimal stimulus intensity, all participants received electrical stimulation to the left cymba conchae at a frequency of 30 Hz, pulse width of 250 μ s, and a 30-second on/30-second off cycle. Stimulation intensity began at 0.4 V and was gradually increased in 0.4 V increments until a tingling sensation was reported, then adjusted to a tolerable, pain-free level, establishing the final stimulus intensity.

The sham stimulation group underwent the same procedure with electrodes placed on the left cymba conchae. The optimal stimulus intensity was confirmed, but the stimulation was turned off after setting the intensity. All patients were informed that they might or might not experience any sensations during the procedure.

Both groups received surgery and anesthesia by local protocols. A preoperative serratus anterior plane block (SAPB) was performed under ultrasound guidance using 20 mL of 0.375% ropivacaine. Postoperative pain was managed using a patient-controlled intravenous analgesia (PCIA) pump, which was programmed to deliver sufentanil (250 μ g/250 mL) at a continuous infusion rate of 1 mL/h, with a bolus dose of 2 mL, a 5-minute lockout interval, and a maximum hourly limit of 12 mL. The duration of PCIA

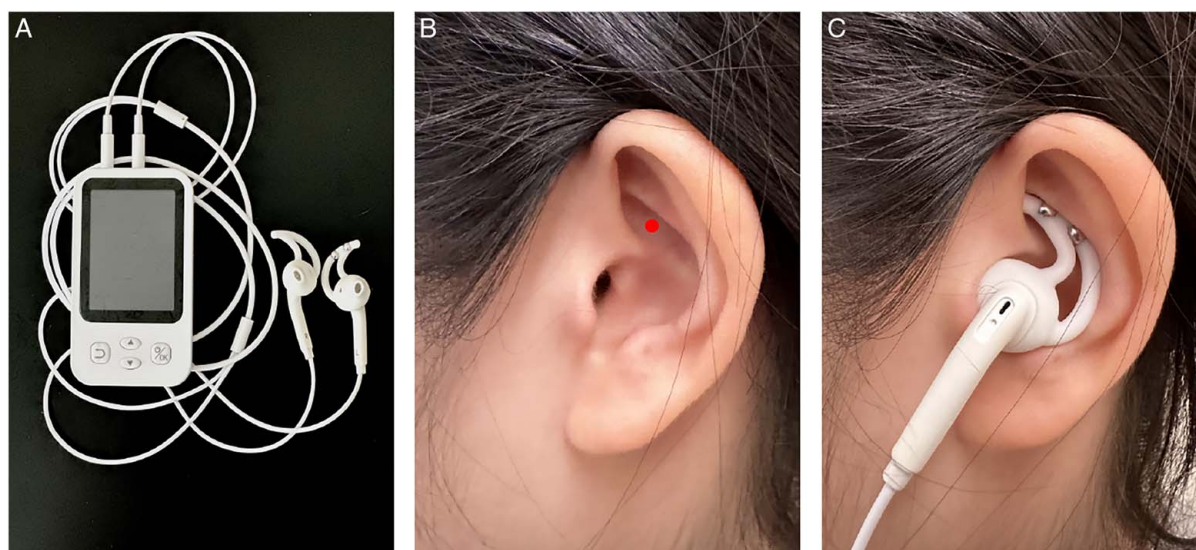


FIGURE 1. Transcutaneous auricular vagus nerve stimulator. Instrument physical appearance. A, Only the left earpiece, equipped with embedded metal electrodes for targeted stimulation of the cymba concha, was used. B, Location of transcutaneous auricular vagus nerve stimulation in the human ear, indicated by the red point. C, Schematic representation of the stimulation apparatus worn by participants in both groups.

use was standardized at 72 hours or until discharge, with no supplementary bolus doses provided by nursing staff.

Primary Outcome

The primary outcome was the cumulative sufentanil consumption during the first 24 hours postoperatively.

Secondary Outcomes

The study evaluated acute pain scores at 6, 12, 24, and 48 hours postoperatively using the numerical rating scale (NRS) for both resting pain and pain during deep breathing. In addition, cumulative sufentanil consumption via the PCIA pump was recorded at these time points.

In the case of rebound pain, it occurred after the anesthetic effects of the peripheral nerve block had worn off. The rebound pain score was calculated as the highest NRS score within 12 hours after blocking wore off minus the lowest NRS score during the effective period. A rebound pain score greater than 0 indicated the presence of rebound pain.²⁰

At 6, 12, and 24 hours postoperatively, during acute pain assessments, patients were asked to report the average NRS score for the preceding 6 hours, along with their minimum and maximum pain levels. This evaluation was essential for identifying and assessing rebound pain.

Regarding the incidence of CPSP and quality of life, patients underwent telephone follow-up at 3 months postoperatively, during which the following questions were asked: (1) whether patients currently had pain related to their chest surgery (yes/no); (2) what was the extent of chronic pain? Average pain intensity over the past week was measured using the NRS (0-10), where 0 indicates no pain and 10 represents the most intense pain; (3) How much the patient's weight had changed recently? And (4) whether the patient's current pain was interfering with sleep (yes/no).

Additional clinical outcomes included the incidence of acute respiratory distress syndrome, pulmonary embolism, stroke, all-cause mortality, and duration of hospital stay. Safety outcomes included adverse events related to taVNS,

such as nausea, vomiting, auricular discomfort, flu-like symptoms, and skin irritation.

Sample Size

In VATS, the SAPB was used for postoperative analgesia, with a mean 24-hour morphine consumption of 42 mg and a SD of 12 mg.² A reduction of 10 mg in 24-hour postoperative opioid consumption was deemed clinically significant,²¹ with a significance level 0.05 and the power 90%, and a 2-sided test. Calculations using G*Power software (version 3.1.9.7) indicated a sample size of 60 was required. Considering an expected dropout rate of ~10%, the study included 70 patients, with 35 patients per group.

Statistical Analysis

All statistical analyses were conducted using SPSS software (version 22.0; IBM SPSS, Chicago, IL). Continuous variables were initially assessed for normality. Data that followed a normal distribution were reported as mean $\pm$ SD and compared using a *t* test. For data that did not follow a normal distribution, values were reported as median with interquartile range (IQR) and analyzed using the Mann-Whitney *U* test. Categorical variables were presented as frequencies (*n*) and percentages (%) and analyzed using Fisher's exact test. A 2-sided *P*-value of less than 0.05 was considered statistically significant.

For the primary outcome, an independent *t* test was conducted to compare the groups. A generalized linear model (GLM) was then applied, adjusting for age, sex, and surgical duration.

For repeated measures in secondary outcomes, including postoperative cumulative sufentanil consumption and NRS pain scores at various time points, a linear mixed-effects model (LMM) assuming an autoregressive correlation structure was fitted separately for each parameter. This model included both a patient-level random effect to account for within-patient correlation and fixed effects for treatment, time, and their interaction. The treatment-by-

time interaction was evaluated with a significance threshold of 0.20. If significant, the treatment effect was reported separately for each time point with Bonferroni correction. Otherwise, the treatment effect was assessed across all time points by collapsing the data and removing the interaction term from the model.

RESULTS

Between March 1, 2024, and July 9, 2024, 1,059 patients were screened for eligibility, and 70 participants were enrolled and randomized. Of these, 3 patients required conversion to open surgery due to pleural adhesions, resulting in 67 participants who completed the study (Fig. 2).

The baseline characteristics of the randomized participants are presented in Table 1. The mean age (SD) was 67 (7) years for the stimulation group and 64 (8) years for the sham group. Gender distribution was 22 males (65%) in the stimulation group and 17 males (52%) in the sham stimulation group.

The primary outcome of sufentanil consumption during the first 24 hours after surgery did not differ significantly between the 2 groups. The mean (SD) sufentanil consumption was 33 μ g (13) in the active group and 32 μ g (9) in the sham group, with an unadjusted mean difference = 1.2 (95% CI = -4.2 to 6.6, unadjusted *P*-value = 0.661) (Table 2). After adjusting for age, gender, and surgical duration using a GLM, the analysis revealed an

adjusted mean difference of 2.1 (95% CI = -3.6 to 7.7), with an adjusted *P*-value of 0.561.

For repeated measures of secondary outcomes, our primary analysis showed no statistically significant time-by-treatment interaction for postoperative sufentanil consumption (*P*-value = 0.208), NRS resting pain (*P*-value = 0.719), or NRS pain during deep breathing (*P*-value = 0.627). Therefore, overall group differences were calculated across all time points by collapsing the data, excluding the interaction term. No statistically significant differences in sufentanil consumption were observed between the groups (*P*-value = 0.770), with overall group differences across all time points (6, 12, 24, and 48 h postoperatively) evaluated by aggregating the data. Compared with the sham stimulation group, taVNS significantly decreased the NRS score for resting pain (*P*-value = 0.001) and deep breathing pain (*P*-value < 0.001) postoperatively (Fig. 3).

Furthermore, taVNS was associated with a significant reduction in the incidence of rebound pain following SAPB (Table 3). The prevalence of rebound pain decreased from 58% to 18% (odds ratio [OR] = 0.18, 95% CI = 0.05 to 0.46, *P*-value = 0.001). In the active stimulation group, the median (IQR) NRS score for rebound pain was 2 (1 to 2), compared with 2 (1 to 3) in the sham stimulation group.

Fewer patients in the active stimulation group complained of CPSP at the 3-month follow-up. The incidence of CPSP decreased from 27% to 6% (OR = 0.17, 95% CI = 0.04-0.76, *P*-value = 0.018). In the sham stimulation group, 9 patients experienced CPSP, with pain scores having a median (IQR) of 3 (2 to 5), and 3 of these patients reported

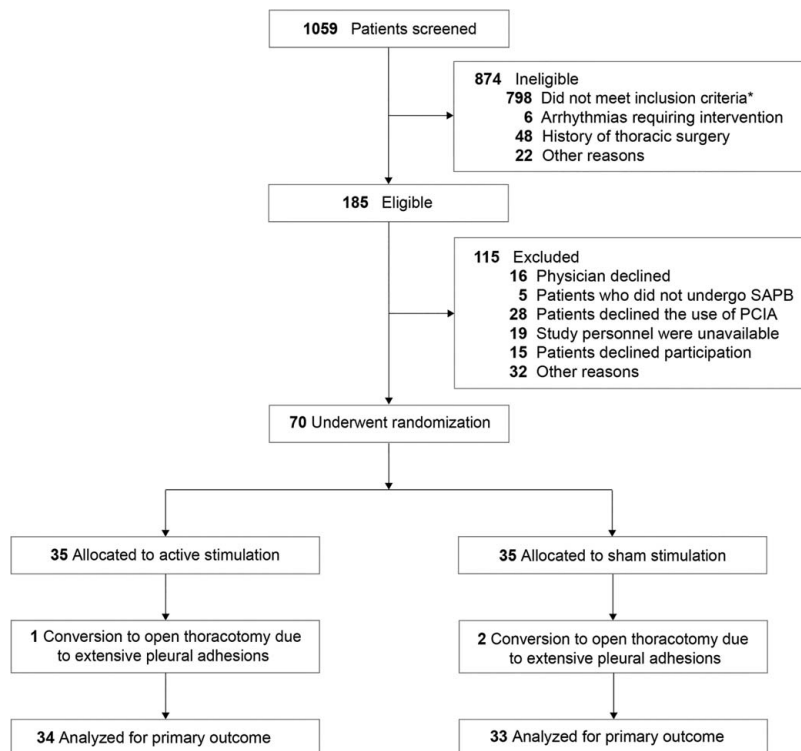


FIGURE 2. Recruitment, randomization, and follow-up in the trial. Abbreviations: ASA, American Society of Anesthesiologists; VATS, video-assisted thoracoscopic surgery; SAPB, serratus anterior plane block; PCIA, patient-controlled intravenous analgesia. *Of the 783 patients excluded due to surgery type, 662 underwent VATS wedge or pulmonary segment resection for lung cancer, 73 underwent mediastinal tumor resections, 15 underwent bullous lung resections, 15 underwent bilateral sympathectomies, and 18 underwent other procedures. In addition, 7 patients were classified as ASA IV, and 8 were over 85 years old.

TABLE 1. Baseline Characteristics of the Randomized Patients

	Active group (n = 35)	Sham group (n = 35)	Standardized difference*
No. patients with available data, n	34	33	
Age (y), mean (SD)	67 (7)	64 (9)	0.372
Sex, n (%)			
Male	22 (65)	17 (52)	0.266
Female	12 (35)	16 (48)	
Body mass index, mean (SD), kg/m ²	23 (3)	24 (3)	0.333
Medical history, n (%)			
Hypertension	16 (47)	16 (49)	0.040
Smoking	10 (29)	9 (27)	0.045
Diabetes	7 (21)	7 (21)	0.012
Coronary artery disease	5 (15)	5 (15)	0.013
Pneumonia	5 (15)	4 (12)	0.088
Stroke	2 (6)	0	NA
ASA physical status, n (%)			0.244
II	27 (79)	29 (88)	
III	7 (21)	4 (12)	
Single-lung ventilation, n (%)			0.159
Double	19 (56)	21 (64)	
Bronchial blocker	15 (44)	12 (36)	
Duration of surgery, median (IQR), minutes	112 (70-157)	100 (78-135)	0.220

*Standardized differences (Cohen's d) were computed by dividing group differences by the pooled SD.
ASA indicates American Society of Anesthesiologists; IQR, interquartile range; NA, not analyzed.

sleep disturbances related to CPSP. In the active stimulation group, 2 patients experienced CPSP, with NRS scores of 3 and 5, respectively, and 1 patient reported sleep disturbances. Notably, 1 patient was lost to follow-up.

The usage of taVNS, including stimulation intensity (both adapted and actual), duration of stimulation, and postoperative utilization, was recorded (Table 4). No adverse events related to taVNS were reported.

Supplementary table, Supplemental Digital Content 2, <http://links.lww.com/CJP/B220> summarized the perioperative and recovery outcomes, indicating no significant differences between the active and sham stimulation groups in terms of intraoperative sufentanil dose, extubation time, PACU length of stay, or hospital length of stay. One patient in the active stimulation group developed dysphasia, left-sided ptosis, and left-hand weakness. These symptoms occurred 6 hours after the administration of the intervention. An emergency head computed tomography (CT) scan was unremarkable, and a transient ischemic attack was diagnosed.

DISCUSSION

The taVNS did not significantly reduce cumulative opioid consumption within the 24 hours after thoracoscopic

lobectomy. However, it was associated with significant reductions in several secondary outcomes, including postoperative NRS scores for acute pain, rebound pain following SAPB, and the incidence of CPSP within 3 months.

The taVNS did not reduce opioid consumption postoperatively. Possible reasons include the administration of an ultrasound-guided SAPB before surgery, which significantly lowered postoperative opioid consumption,²² and the use of a postoperative PCIA pump programmed with a continuous background dose to minimize pain-associated sleep disturbances. In addition, placebo effects may have been amplified because the sham device was configured similarly to the active device, leading patients to expect a treatment effect.²³ This may account for the comparable postoperative opioid consumption in both groups, with 33 mg in the active group and 32 mg in the sham group (all doses converted to intravenous morphine milligram equivalents), both lower than the 42-58 mg reported in similar studies.^{2,24} At our center, PCIA pump guidelines recommend administering bolus doses only when the NRS pain score reaches ≥ 3. As both groups received nerve blocks before surgery, their pain scores were generally lower. Consequently, bolus use was infrequent, as evidenced by the

TABLE 2. Primary Outcome of the Study

	Active group (n = 35)	Sham group (n = 35)	Mean difference (95% CI)		P	
			Unadjusted	Adjusted	Unadjusted	Adjusted
No. patients with available data, n	34	33	–	–	–	–
Sufentanil consumption*, mean (SD), µg	33 (13)	32 (9)	1.2 (–4.2 to 6.6)†	2.1 (–3.7 to 7.7)‡	0.661†	0.561‡

*During the first 24 hours after surgery.
†An independent *t* test was used to compare the groups.
‡A generalized linear model (GLM) was used after adjusting for age, gender, and surgical duration.
CI indicates confidence interval.

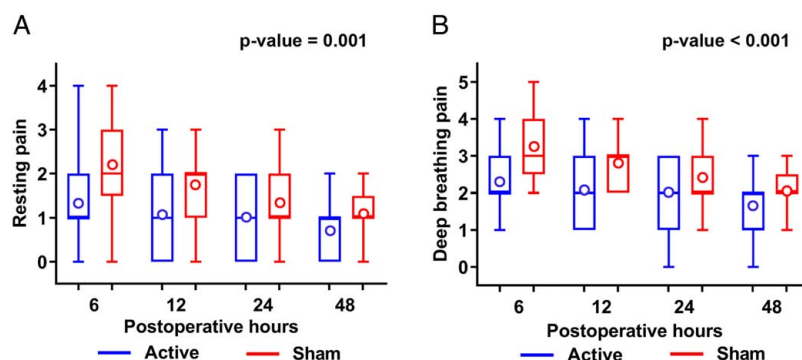


FIGURE 3. Numerical rating scale (NRS) of postoperative pain within 48 hours. Pain of resting (A) and deep breathing states (B). Data are presented as median (horizontal bars), interquartile ranges (box), whiskers (min to max), and mean (bold circle). A linear mixed-effects model (LMM) with an autoregressive correlation structure assessed overall group differences across all time points by aggregating NRS scores for resting pain and pain during deep breathing, excluding the interaction term, as the treatment-by-time interaction exceeded the 0.20 significance threshold.

cumulative opioid consumption at each time point. This may have been one of the key factors contributing to the negative result of the primary outcome.

Compared with the sham stimulation group, patients in the active stimulation group had significantly lower NRS pain scores for resting pain and deep breathing pain postoperatively, which is consistent with previous experimental findings.^{17–19} This suggests that taVNS may have potential as a noninvasive approach for perioperative pain management. In our study, the use of nerve blocks combined with continuous PCIA background infusion

may have masked potential differences between the active and sham stimulation groups. Although statistically significant differences in pain scores were observed, the magnitude of these differences was relatively small, and their clinical significance remains uncertain. Future studies, such as those excluding nerve blocks or PCIA background infusion, may provide a clearer evaluation of the effects of taVNS on pain relief.

The taVNS was effective in reducing the occurrence of rebound pain following nerve block. Previous studies have shown that 40% to 50% of patients experience rebound

TABLE 3. Secondary Outcomes of the Study

	Active group (n = 35)	Sham group (n = 35)	Effect estimate (95% CI)	P
No. patients with available data, n	34	33		
Sufentanil consumption postoperatively, median (IQR), μg				0.559*
6 h	10 (7)	10 (5)	−0.41 (−3.32 to 2.50)	
12 h	17 (9)	17 (6)	−0.02 (−3.65 to 3.60)	
24 h	33 (13)	32 (9)	1.20 (−4.25 to 6.65)	
48 h	56 (28)	51 (17)	4.64 (−6.33 to 16.22)	
NRS for resting pain postoperatively, median (IQR)				0.001†
6 h	1 (1–2)	2 (2–3)	−0.86 (−1.35 to −0.37)	
12 h	1 (0–2)	2 (1–2)	−0.64 (−1.03 to −0.25)	
24 h	1 (0–2)	1 (1–2)	−0.33 (−0.71 to 0.05)	
48 h	1 (0–1)	1 (1–2)	−0.39 (−0.73 to −0.04)	
NRS for deep breathing pain postoperatively, median (IQR)				<0.001†
6 h	2 (2–3)	3 (3–4)	−0.95 (−1.42 to −0.48)	
12 h	2 (1–3)	3 (2–3)	−0.70 (−1.08 to −0.32)	
24 h	2 (1–3)	2 (2–3)	−0.39 (−0.78 to −0.01)	
48 h	2 (1–2)	2 (2–3)	−0.41 (−0.78 to −0.04)	
Rebound pain, n (%)	6 (18)	19 (58)	0.18 (0.05 to 0.46)	0.001‡
3-month follow-up, n	33	33		
CPSP, n (%)	2 (6)	9 (27)	0.17 (0.04 to 0.76)	0.018‡
NRS, median (IQR)	0 (0–0)	0 (0–2)	NA	NA
Change of weight, median (IQR), kg	0 (0–0)	0 (−3–0)	NA	NA
Change of sleep, n (%)	1 (3)	3 (9)	NA	NA
Missing, n (%)	1 (3)	0	NA	NA

*A linear mixed-effects model (LMM) with an autoregressive correlation structure was used to evaluate the treatment effect by collapsing the data over time and excluding the interaction term, as the treatment-by-time interaction exceeded the 0.20 significance threshold.

†Before performing an LMM analysis with an autoregressive correlation structure, the data were ln-transformed.

‡ χ^2 test used to compare proportions between 2 groups.

CI indicates confidence interval; CPSP, chronic postsurgical pain; IQR, interquartile range; NA, not analyzed; NRS, numerical rating scale.

TABLE 4. Patient Adherence

	Active group (n = 35)	Sham group (n = 35)	P*
No. patients with available data, n	34	33	
Stimulation intensity, median (IQR), V			
Adapted intensity	8.0 (8.0-8.8)	8.4 (7.6-9.2)	0.879
Actual intensity	8.0 (8.0-8.8)	—	NA
On surgery day, n (%)	34 (100)	33 (100)	NA
Stimulation duration, median (IQR), minutes	195 (158-270)	180 (165-233)	0.620
Postoperative day 1, n (%)	33 (97)	31 (94)	0.614
Postoperative day 2, n (%)	24 (71)	22 (67)	0.796

*Mann-Whitney *U* test used to compare medians and χ^2 test used to compare proportions.

IQR indicates interquartile range; NA, not analyzed.

pain.²⁵⁻²⁷ This type of pain can affect recovery and increase the demand for medical resources. Our study found that the incidence of rebound pain in the sham stimulation group was 58%, while the stimulation group experienced a statistically significant reduction to 18%. This reduction may also have clinical significance, warranting further investigation.

There is a strong relationship between pain and quality of life. The taVNS, as an innovative, effective, and noninvasive method for the prevention of postoperative CPSP and improvement of quality of life, holds significant potential and promise. The taVNS reduced the incidence of CPSP at 3 months post-thoracoscopic lobectomy, with rates of 27% in the sham stimulation group and 6% in the active stimulation group. As is well known, CPSP affects 25% to 44% of thoracic surgery patients, significantly impacting their quality of life.^{3,5-7} Although sleep disturbances and weight reduction were more common in the sham group, possibly linked to CPSP, the small sample size precluded statistical analysis. The cause of these changes remains unclear and warrants further investigation.

Our study had several limitations. Although previous research has demonstrated that cholinergic nerves have anti-inflammatory^{28,29} and analgesic properties,^{9,17-19} as well as evidence that taVNS can achieve effects like invasive vagus nerve stimulation,³⁰ our study only assessed subjective pain measures and did not evaluate objective indicators of autonomic nervous system function, such as heart rate variability, sympathetic tone, or vagal tone. Moreover, although evidence suggests that perioperative vagal stimulation may enhance postoperative recovery,⁸ our evaluation of recovery using the quality of recovery-15 (QOR-15) scale was constrained by poor patient adherence. However, patients in this study were followed for 3 months, with assessments of pain, weight, and sleep, which may provide insights into the potential long-term impact of taVNS on quality of life.

The taVNS may have a potential role in managing both short-term and long-term postoperative pain. Effective postoperative pain management not only alleviates acute pain but may also reduce the risk of CPSP, making proactive pain management strategies crucial following thoracic surgery. However, no corrections for multiple comparisons were applied to the secondary outcomes in this study, potentially increasing the risk of type I error. Accordingly, these findings are preliminary and require

further investigation. Future research should investigate the effects of taVNS across different surgical procedures and patient populations and evaluate its potential to regulate the autonomic nervous system for a better understanding of its impact on postoperative recovery.

CONCLUSION

The taVNS did not significantly reduce postoperative opioid consumption in patients undergoing thoracoscopic lobectomy. However, it was associated with improvements in lower postoperative acute pain scores, reduced incidences of rebound pain following nerve block, and decreased prevalences of CPSP at 3 months postoperatively. These findings require further confirmation in subsequent investigations and may guide future clinical practice in this area.

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REFERENCES

- McCall P, Steven M, Shelley B. Anaesthesia for video-assisted and robotic thoracic surgery. *BJA Educ*. 2019;19:405-411.
- Kim DH, Oh YJ, Lee JG, et al. Efficacy of ultrasound-guided serratus plane block on postoperative quality of recovery and analgesia after video-assisted thoracic surgery: a randomized, triple-blind, placebo-controlled study. *Anesth Analg*. 2018;126:1353-1361.
- Zhang Y, Zhou R, Hou B, et al. Incidence and risk factors for chronic postsurgical pain following video-assisted thoracoscopic surgery: a retrospective study. *BMC Surg*. 2022;22:76.
- Sort R, Brorson S, Gögenur I, et al. Peripheral nerve block anaesthesia and postoperative pain in acute ankle fracture surgery: the ankle randomised trial. *Br J Anaesth*. 2021;126:881-888.
- Moorthy A, NE A, Dempsey E, et al. Postoperative recovery with continuous erector spinae plane block or video-assisted paravertebral block after minimally invasive thoracic surgery: a prospective, randomised controlled trial. *Br J Anaesth*. 2023;130:e137-e147.
- Chen WC, Bai YY, Zhang LH, et al. Prevalence and predictors of chronic postsurgical pain after video-assisted thoracoscopic surgery: a systematic review and meta-analysis. *Pain Ther*. 2023;12:117-139.
- Bayman EO, Parekh KR, Keech J, et al. A prospective study of chronic pain after thoracic surgery. *Anesthesiology*. 2017;126:938-951.
- Patel ABU, Weber V, Gourine AV, et al. The potential for autonomic neuromodulation to reduce perioperative complications and pain: a systematic review and meta-analysis. *Br J Anaesth*. 2022;128:135-149.
- Sullere S, Kunczt A, McGehee DS. A cholinergic circuit that relieves pain despite opioid tolerance. *Neuron*. 2023;111:3414-3434.e15.
- Wang X, Huang ZG, Gold A, et al. Propofol modulates gamma-aminobutyric acid-mediated inhibitory neurotransmission to cardiac vagal neurons in the nucleus ambiguus. *Anesthesiology*. 2004;100:1198-1205.
- May SM, Chiang E, Reyes A, et al. Neuromodulation of innate immunity by remote ischaemic conditioning in humans: experimental cross-over study. *Brain Behav Immun Health*. 2021;16:100299.

12. Ackland GL, Abbott TEF, Cain D, et al. Preoperative systemic inflammation and perioperative myocardial injury: prospective observational multicentre cohort study of patients undergoing non-cardiac surgery. *Br J Anaesth*. 2019;122:180–187.
13. May SM, Reyes A, Martir G, et al. Acquired loss of cardiac vagal activity is associated with myocardial injury in patients undergoing noncardiac surgery: prospective observational mechanistic cohort study. *Br J Anaesth*. 2019;123:758–767.
14. Abbott TEF, Pearse RM, Cuthbertson BH, et al. Cardiac vagal dysfunction and myocardial injury after non-cardiac surgery: a planned secondary analysis of the measurement of exercise tolerance before surgery study. *Br J Anaesth*. 2019;122:188–197.
15. Zhang J, Shen QH, Chen G. Effect of transauricular nerve stimulation on perioperative pain: a single-blind, analyser-masked, randomised controlled trial. Comment on Br J Anaesth 2023; 130: 468–76. *Br J Anaesth*. 2025;134:579–580.
16. Badran BW, Brown JC, Dowdle LT, et al. Tragus or cymba conchae? Investigating the anatomical foundation of transcutaneous auricular vagus nerve stimulation (tavns). *Brain Stimul*. 2018;11:947–948.
17. Patel ABU, Bibawy P, Althonayan JIM, et al. Effect of transauricular nerve stimulation on perioperative pain: a single-blind, analyser-masked, randomised controlled trial. *Br J Anaesth*. 2023;130:468–476.
18. Ilfeld BM, Finneran JJ, Alexander B, et al. Percutaneous auricular neuromodulation (nerve stimulation) for the treatment of pain following total knee arthroplasty: a randomized, double-masked, sham-controlled pilot study. *Reg Anesth Pain Med*. 2025;50:26–35.
19. Ilfeld BM, Abramson WB, Alexander B, et al. Percutaneous auricular neuromodulation (nerve stimulation) for the treatment of pain following cholecystectomy and hernia repair: a randomized, double-masked, sham-controlled pilot study. *Reg Anesth Pain Med*. 2024;49:628–634.
20. Williams BA, Bottegal MT, Kentor ML, et al. Rebound pain scores as a function of femoral nerve block duration after anterior cruciate ligament reconstruction: retrospective analysis of a prospective, randomized clinical trial. *Reg Anesth Pain Med*. 2007;32:186–192.
21. Sandeep B, Huang X, Li Y, et al. A comparison of regional anesthesia techniques in patients undergoing video-assisted thoracic surgery: a network meta-analysis. *Int J Surg*. 2022;105:106840.
22. Feray S, Lubach J, Joshi GP, et al. Prospect guidelines for video-assisted thoracoscopic surgery: a systematic review and procedure-specific postoperative pain management recommendations. *Anaesthesia*. 2022;77:311–325.
23. Hróbjartsson A, Gotzsche PC. Placebo interventions for all clinical conditions. *Cochrane Database Syst Rev*. 2010;2010: Cd003974.
24. Finnerty DT, McMahon A, McNamara JR, et al. Comparing erector spinae plane block with serratus anterior plane block for minimally invasive thoracic surgery: a randomised clinical trial. *Br J Anaesth*. 2020;125:802–810.
25. Touil N, Pavlopoulou A, Barbier O, et al. Evaluation of intraoperative ketamine on the prevention of severe rebound pain upon cessation of peripheral nerve block: a prospective randomised, double-blind, placebo-controlled study. *Br J Anaesth*. 2022;128:734–741.
26. Jen TTH, Ke JXC, Wing KJ, et al. Development and internal validation of a multivariable risk prediction model for severe rebound pain after foot and ankle surgery involving single-shot popliteal sciatic nerve block. *Br J Anaesth*. 2022;129:127–135.
27. Barry GS, Bailey JG, Sardinha J, et al. Factors associated with rebound pain after peripheral nerve block for ambulatory surgery. *Br J Anaesth*. 2021;126:862–871.
28. Rosas-Ballina M, Olofsson PS, Ochani M, et al. Acetylcholine-synthesizing t cells relay neural signals in a vagus nerve circuit. *Science*. 2011;334:98–101.
29. Borovikova LV, Ivanova S, Zhang M, et al. Vagus nerve stimulation attenuates the systemic inflammatory response to endotoxin. *Nature*. 2000;405:458–462.
30. Kraus T, Kiess O, Hösl K, et al. Cns bold fmri effects of sham-controlled transcutaneous electrical nerve stimulation in the left outer auditory canal—a pilot study. *Brain Stimul*. 2013;6:798–804.